

Bacterial Morphology

John F. Fisher, MD • 78-slide current deck • shape → structures → cell wall → Gram stain → spores

6 official objectives

1 ScholarRx brick

Gram + vs Gram -

Peptidoglycan

Binary fission

Endospores

AI handout integrated

SOURCE PRIORITY: the current 78-slide Fisher deck controls terminology, examples, images, and emphasis. The user-supplied Lecture 1: Bacterial Morphology and Structure handout is now the secondary enrichment source. It explicitly follows the lecture flow and adds high-yield clinical correlations. Details that are useful but not directly shown on the current deck are labeled Handout-only enrichment.

Official Week 8 Objectives

1. Differentiate Gram-positive from Gram-negative bacteria.
2. Explain Gram stain theory and technique.
3. Identify basic structural features of bacteria.
4. Explain how bacteria divide.
5. Discuss how bacterial spores resist destruction.
6. Outline the biochemical composition of peptidoglycan.

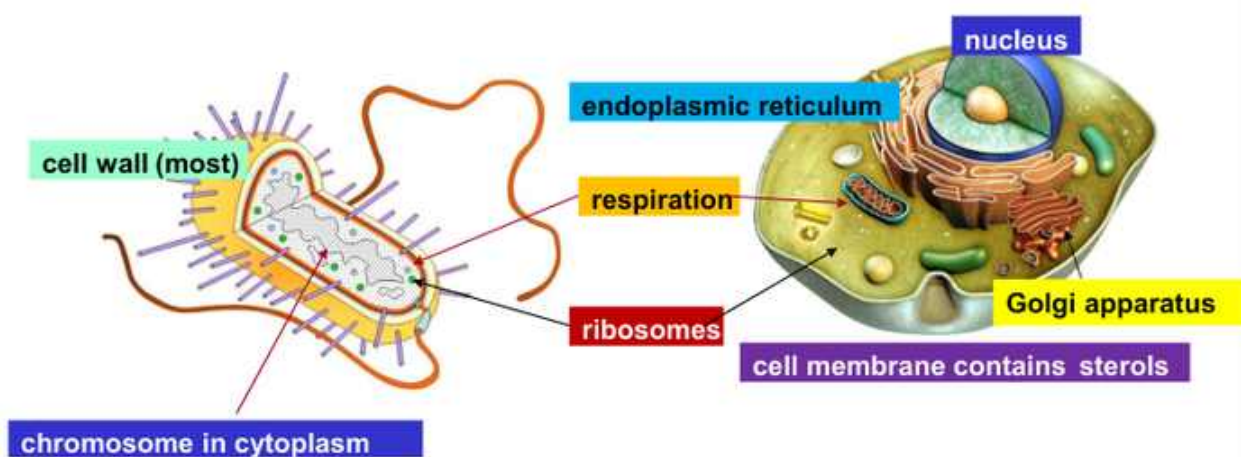
Prokaryote Structure and the Gram Stain

HOW TO STUDY THIS LECTURE

Shape tells you what you are looking at; structure tells you how it behaves; the cell wall explains Gram staining and drug access; spores explain survival.

Slides 2-15: Prokaryotes, Size, Shape & Arrangement

Prokaryotes vs Eukaryotes



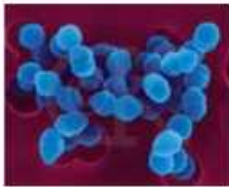
Slide 3. Prokaryote vs eukaryote overview. The bacterial chromosome is in a cytoplasmic nucleoid, most bacteria have a cell wall, and bacterial respiration occurs at the cell membrane rather than mitochondria.

CORE PROKARYOTE PICTURE

No membrane-bound nucleus • nucleoid chromosome • bacterial ribosomes • most have cell wall • cell membrane performs key metabolic functions.

Handout-only clinical connection: bacterial ribosomes are **70S = 30S + 50S**, compared with cytosolic human 80S ribosomes. The handout connects this difference to selective antibiotic targeting by classes such as aminoglycosides and macrolides. It also emphasizes that bacterial membranes usually lack sterols, with **Mycoplasma** as the classic exception.

Coccus (L; G. kokkos, berry)



Bacillus (L. bacillum, little rod)



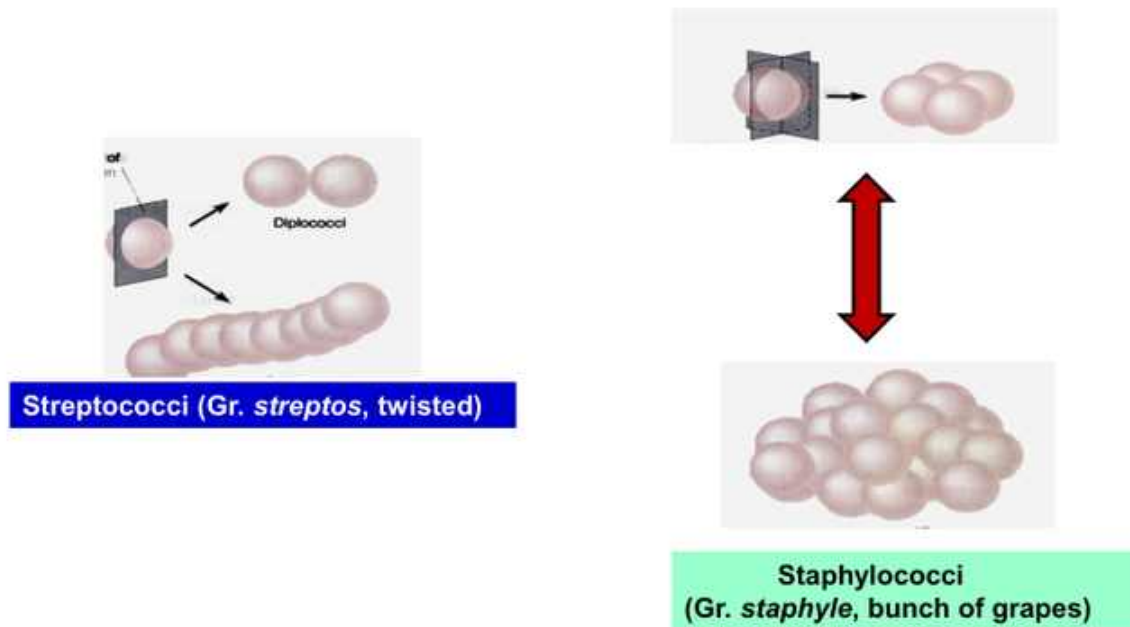
Spiral (L. spira, coil)



Slide 7. Three broad shapes in the deck: coccus, bacillus, and spiral.

Morphology	Deck recognition
Coccus	Spherical/berry-like
Bacillus	Rod-shaped
Spiral	Coiled/helical morphology

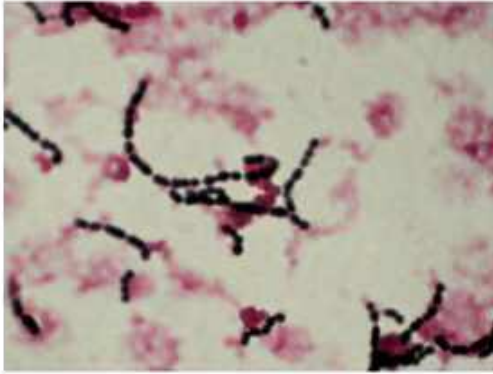
Cocci (planes of cell division)



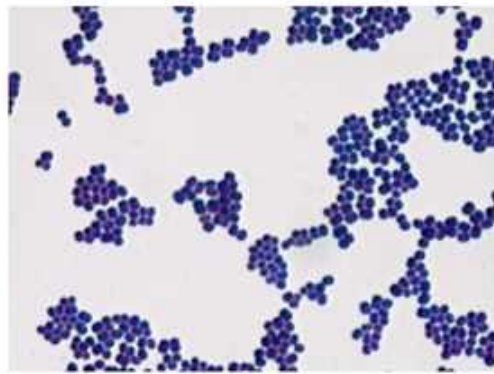
Slide 8. Coccal groupings reflect planes of cell division: chains are streptococcal arrangements; grape-like clusters are staphylococcal arrangements.

Streptococcus vs Staphylococcus

streptococcus



staphylococcus



Slide 9. Direct microscopy comparison of streptococcal chains versus staphylococcal clusters.

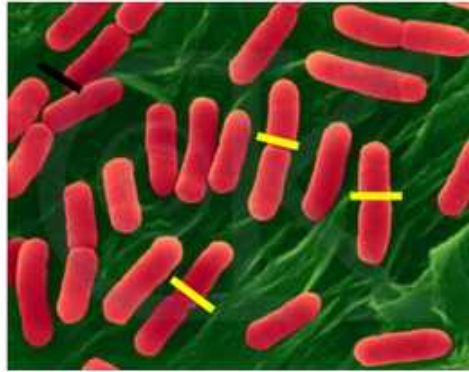
ARRANGEMENT SHORTCUT

Staph = clusters • Strep = chains. Arrangement is a morphology clue, not a full species diagnosis.

Bacilli

(plane of cell division)

across short axis
(fewer groupings)



Slide 10. Bacilli divide across the short axis and therefore show fewer classic grouping patterns than cocci.

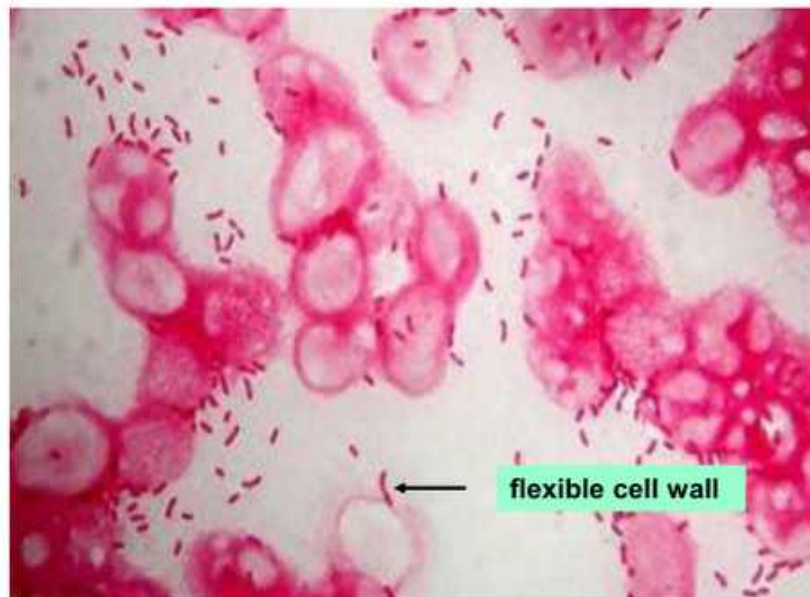
Club-shaped or dumbbell-shaped bacilli



Corynebacterium diphtheriae

Slide 12. *Corynebacterium diphtheriae* is used as the deck example of club- or dumbbell-shaped bacilli.

Curved or comma-shaped bacilli



Vibrio cholerae

Slide 13. *Vibrio cholerae* is the curved/comma-shaped bacillus example.

Spirilla



helical (rigid cell wall; polar flagella)



Helicobacter pylori
(peptic ulcers)



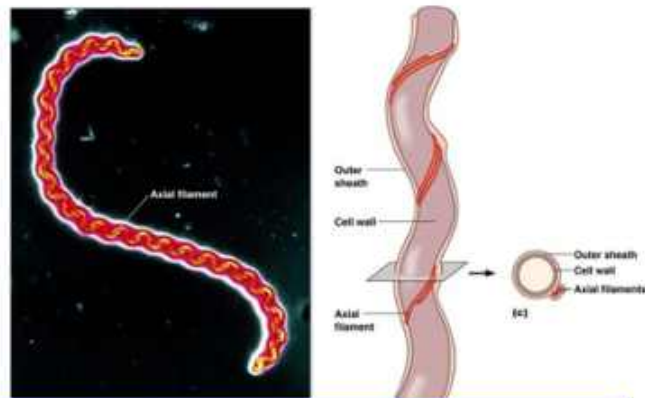
Campylobacter jejuni
(gastroenteritis)



Spirillum minus
(rat bite fever)

Slide 14. Spirilla are rigid helical organisms with polar flagella; examples include *H. pylori*, *Campylobacter jejuni*, and *Spirillum minus*.

Spirochetes



helical (non-rigid cell wall; flagella underneath external sheath)



Borrelia burgdorferi
(Lyme disease)



Leptospira interrogans
(leptospirosis)



Treponema pallidum
(syphilis)

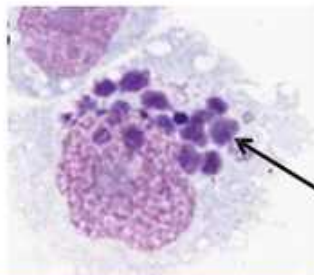
Slide 15. Spirochetes are helical but have a non-rigid wall and flagella beneath an external sheath; examples include *Borrelia*, *Leptospira*, and *Treponema*.

Spirillum vs spirochete: the lecture distinguishes them by rigidity and flagellar location. **Spirilla** = rigid helix + polar flagella. **Spirochetes** = flexible/non-rigid + axial filaments beneath an external sheath.

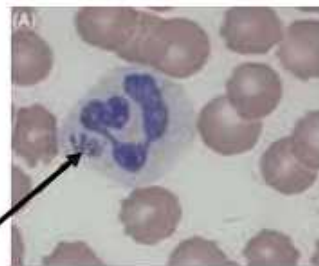
Slides 16-19: Special Bacterial Forms

Anaplasmataceae

obligate intracellular bacteria
(reproduce in morulae)



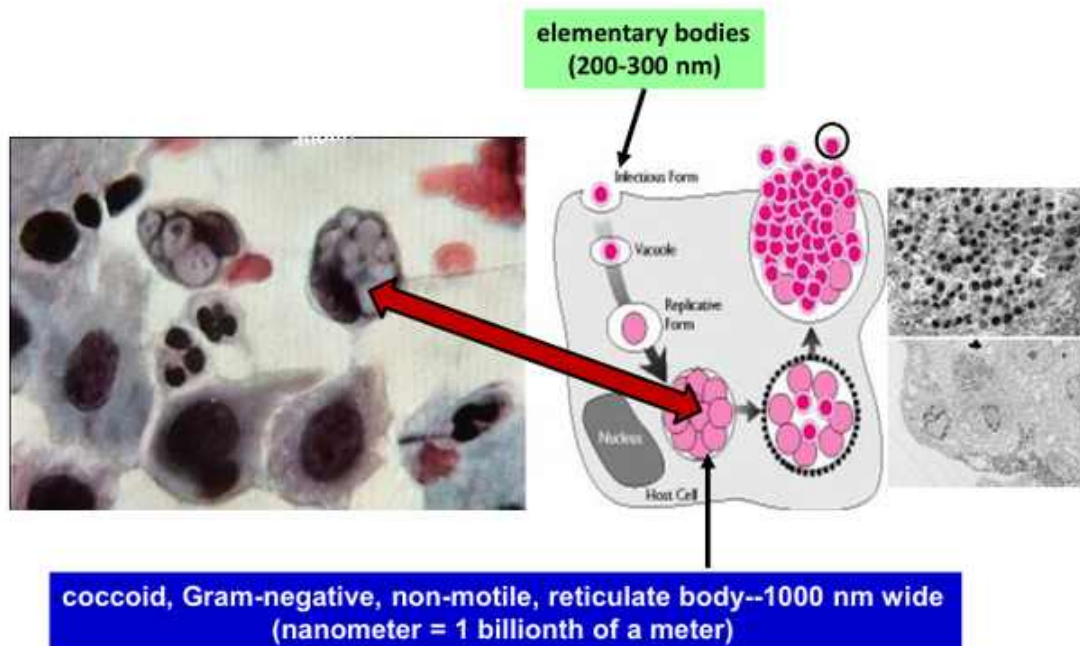
Ehrlichia chaffeensis
(human monocytic anaplasmosis)



Anaplasma phagocytophilum
(human granulocytic anaplasmosis)

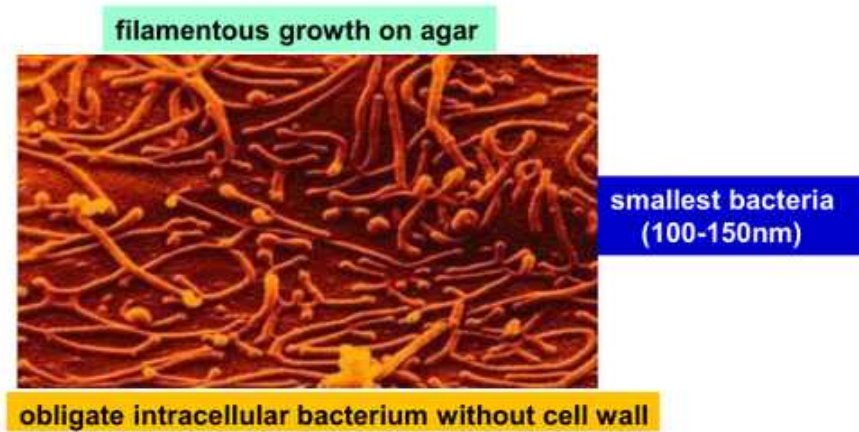
Slide 17. Ehrlichia and Anaplasma are shown reproducing within intracellular morulae; the deck labels the monocyte-vs-granulocyte associations.

Chlamydia & Chlamydophila



Slide 18. Chlamydia/Chlamydophila are shown as coccoid Gram-negative nonmotile organisms with elementary and reticulate-body forms.

Mycoplasma



Slide 19. The current slide labels Mycoplasma as a very small bacterium without a cell wall and shows filamentous growth on agar.

Deck wording note: this companion preserves the instructor slide wording for special organisms rather than silently substituting outside microbiology detail. For course study, know the visual/structural distinctions the deck emphasizes: intracellular forms, elementary/reticulate bodies, and the lack of a cell wall in Mycoplasma.

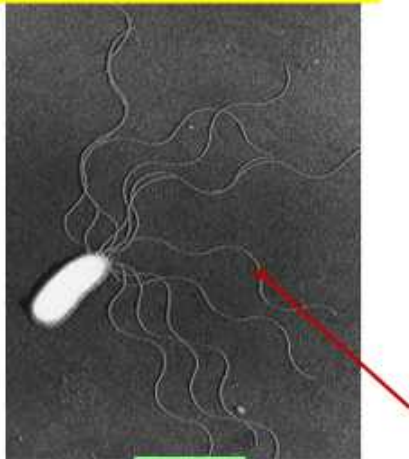
Handout-only clinical enrichment: **Mycoplasma** has no cell wall and stabilizes its membrane with host-derived sterols, so beta-lactam cell-wall agents such as penicillin are ineffective. The handout also contrasts **Mycobacterium**, which has a waxy mycolic-acid-rich envelope and is identified with an acid-fast stain rather than a routine Gram stain.

Slides 20-31, 41-42, 55-56: Basic Bacterial Structures

Flagella = motility

Flagella

composed of flagellin
(protein → hollow cylinder)



E. coli



50 μ / second
(1.1 x 10⁻⁴ mph)

composed of protein → bone,
muscle, ligaments, tendons



Usain Bolt
(100m: 9.58 sec)



27.8 mph at top speed

Slide 21. Flagella are composed of flagellin and form a hollow protein cylinder; the lecture emphasizes bacterial motility.

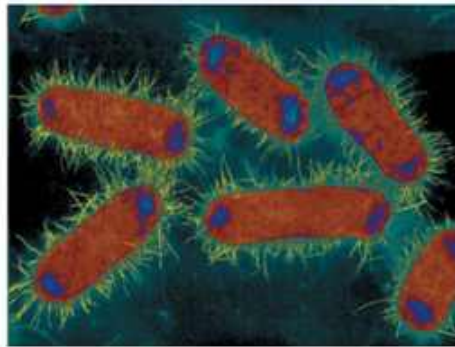
FLAGELLUM ANATOMY

Filament → **hook** → **basal body**. The basal body anchors the flagellum in the cell envelope.

Fimbriae = adherence

Fimbriae

Peritrichous
("fringe")



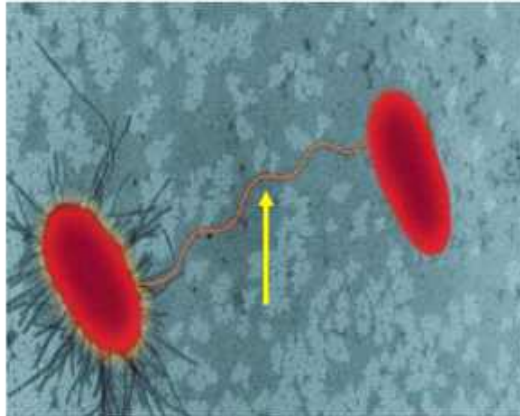
Mediate adherence (surfaces, target cells, one another)

Slide 25. Fimbriae form a fringe-like surface and mediate adherence to surfaces, target cells, and other bacteria.

Fimbriae = stick. They are an adherence structure, not the primary motility structure.

Sex pili = DNA transfer interface

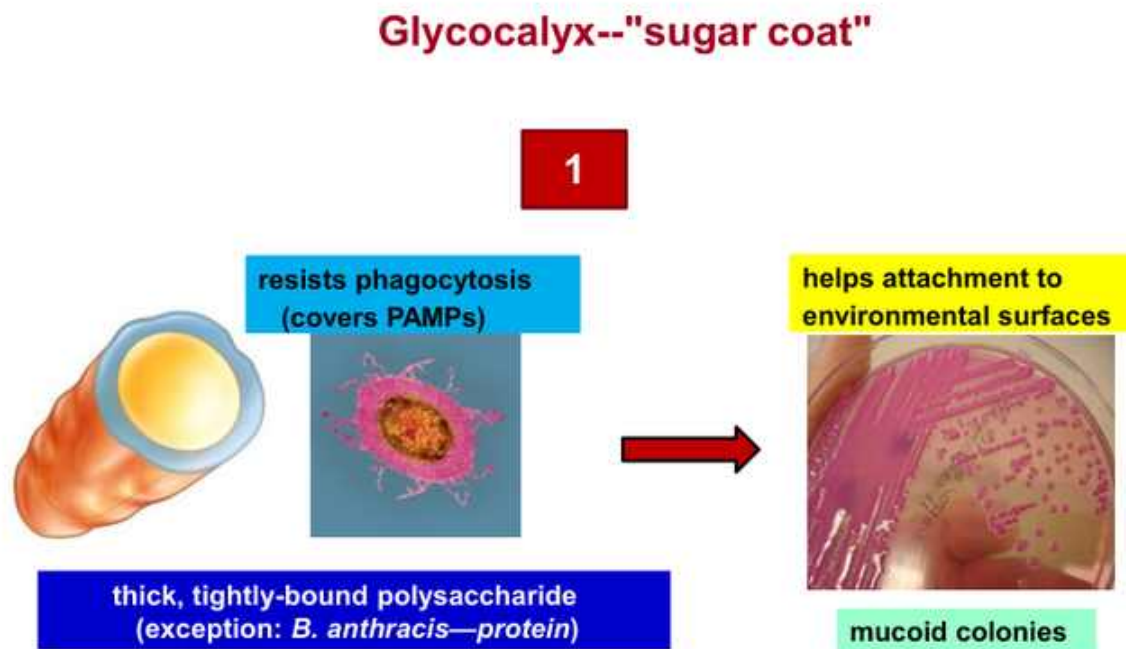
**F pili--fertility factor
(aka sex pili)**



Gram-negative organisms

Slide 27. The deck identifies F pili / sex pili as fertility-factor structures in Gram-negative organisms.

Glycocalyx = protection + attachment + biofilm

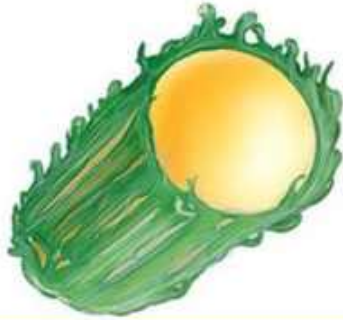


Slide 29. The glycocalyx or sugar coat resists phagocytosis by covering PAMPs and helps attachment. The deck notes most capsules are polysaccharide, with *B. anthracis* as a protein exception.

Glycocalyx

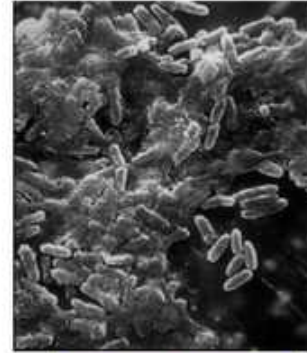
2

"loose" surface attachment



slime layer: polysaccharides, glycoproteins, glycolipids

precursor

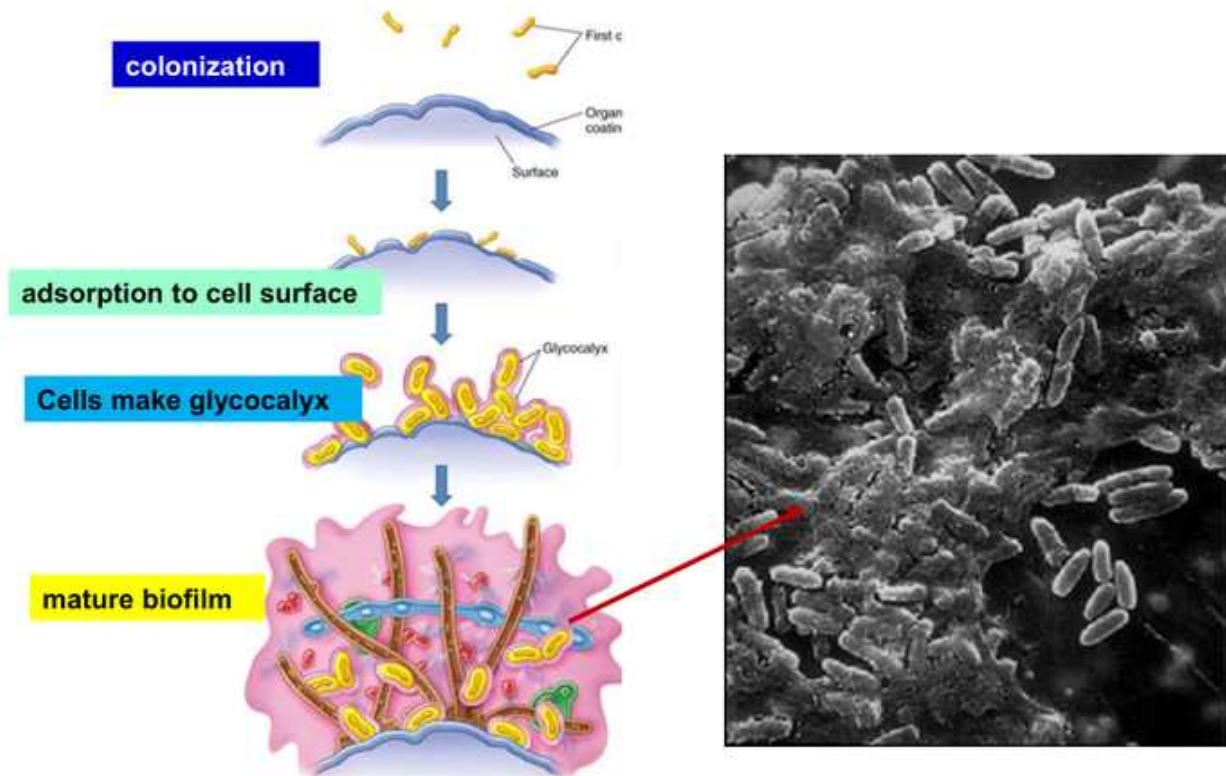


mature biofilm

protects from drying, loss of nutrients, chemical insults (eg antibiotics)

Slide 30. A looser slime layer protects against drying, nutrient loss, and chemical insults and is a precursor to mature biofilm.

Bacteria (biofilm production)



Slide 31. Biofilm sequence in the deck: colonization → adsorption to surface → glycocalyx production → mature biofilm.

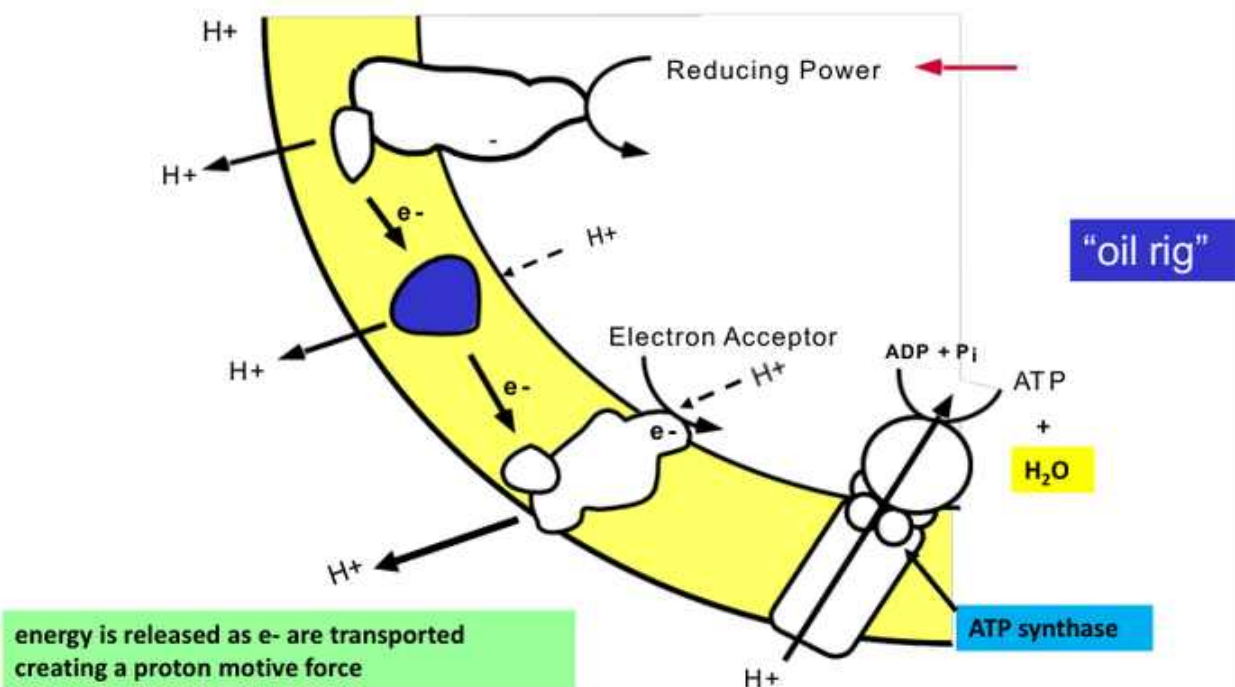
Structure	Deck function
Flagella	Motility
Fimbriae	Adherence
F/sex pili	Fertility/conjugation structure
Capsule / glycocalyx	Anti-phagocytic shielding + adherence
Slime layer	Loose attachment, protection, biofilm precursor
70S ribosome	30S + 50S bacterial ribosomal subunits

Handout-only clinical connections:

- **Plasmids** are small circular extrachromosomal DNA molecules that commonly carry antibiotic-resistance or virulence genes and can move between bacteria during conjugation.

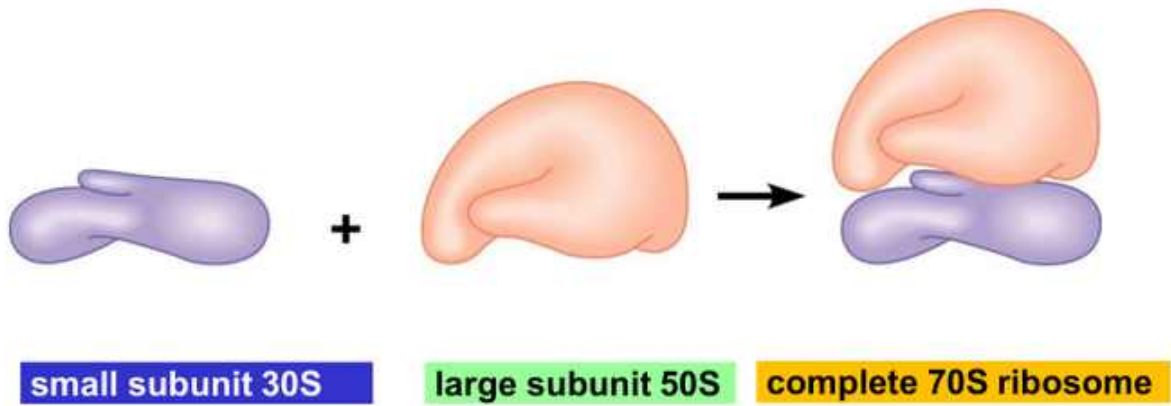
- **Fimbriae** mediate adherence to host surfaces; the handout uses *Neisseria gonorrhoeae* adherence as an example.
- The **sex pilus** is specialized for DNA transfer.
- The **capsule** is organized and antiphagocytic, while a **slime layer** is looser and promotes attachment/biofilm formation.
- The flagellar basal-body motor is powered by a **proton gradient**.

Bacterial cell membrane (site of respiration)



Slide 41. Bacterial cell membrane is the site of respiration in the deck: electron transport generates proton motive force that powers ATP synthase.

Bacterial ribosomes



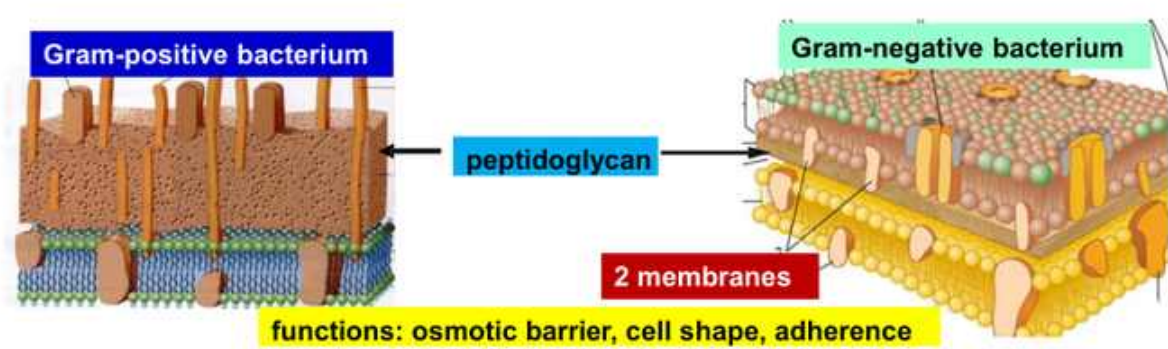
EUKARYOTIC CELL



Slide 56. Bacterial ribosome: 30S small subunit + 50S large subunit = 70S complete ribosome.

Slides 32-40: Cell Wall, Peptidoglycan & Gram Architecture

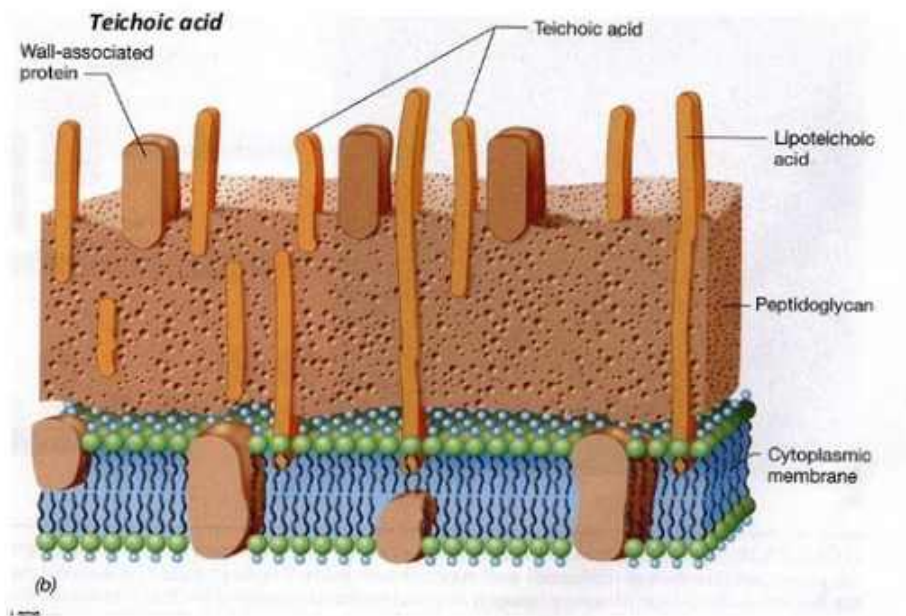
Cell wall--big picture



Slide 33. Big-picture comparison: Gram-positive bacteria have thick peptidoglycan and one membrane; Gram-negative bacteria have a thin peptidoglycan layer plus two membranes.

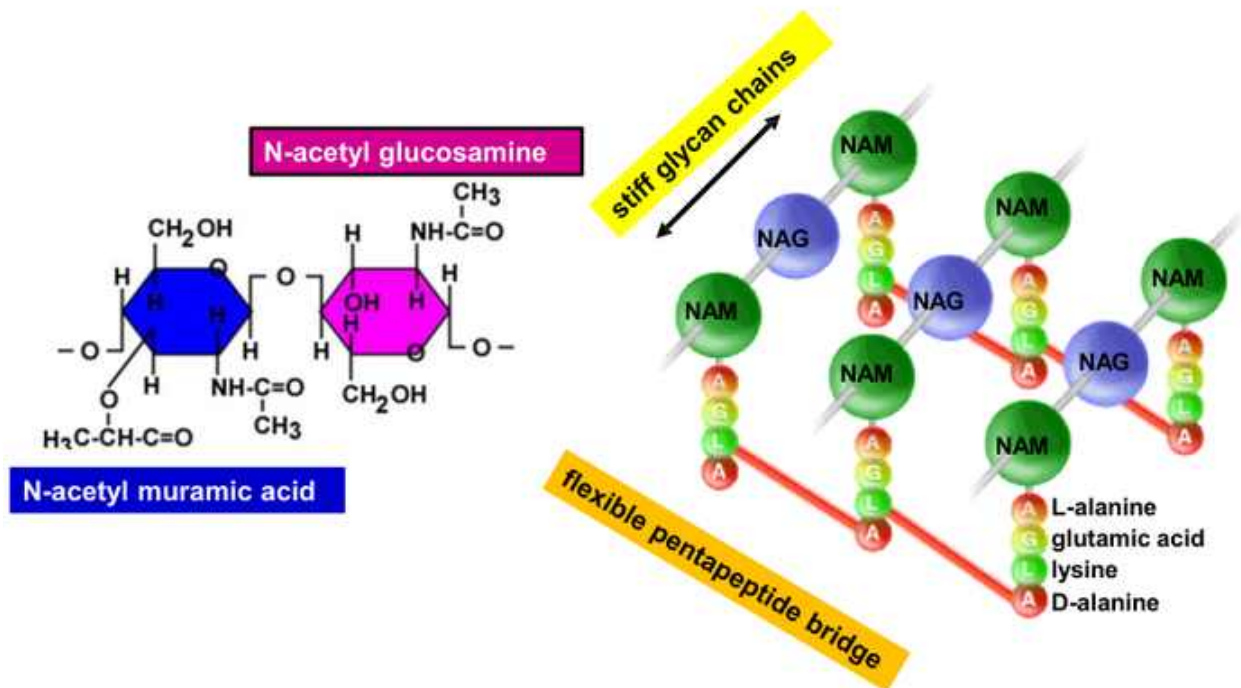
Feature	Gram-positive	Gram-negative
Peptidoglycan	Thick	Thin
Membranes	One cytoplasmic membrane	Inner + outer membrane
Teichoic acids	Present	Absent
LPS	Absent	Outer membrane
Porins	Not the defining feature	Outer-membrane channels
Periplasm	Not emphasized as a large compartment	Present

Gram-positive cell wall



Slide 34. Gram-positive cell wall image: thick peptidoglycan with wall teichoic acid and lipoteichoic acid over the cytoplasmic membrane.

Peptidoglycan



Slide 35. Peptidoglycan chemistry: alternating NAG and NAM glycan chains with peptide side chains and cross-links.

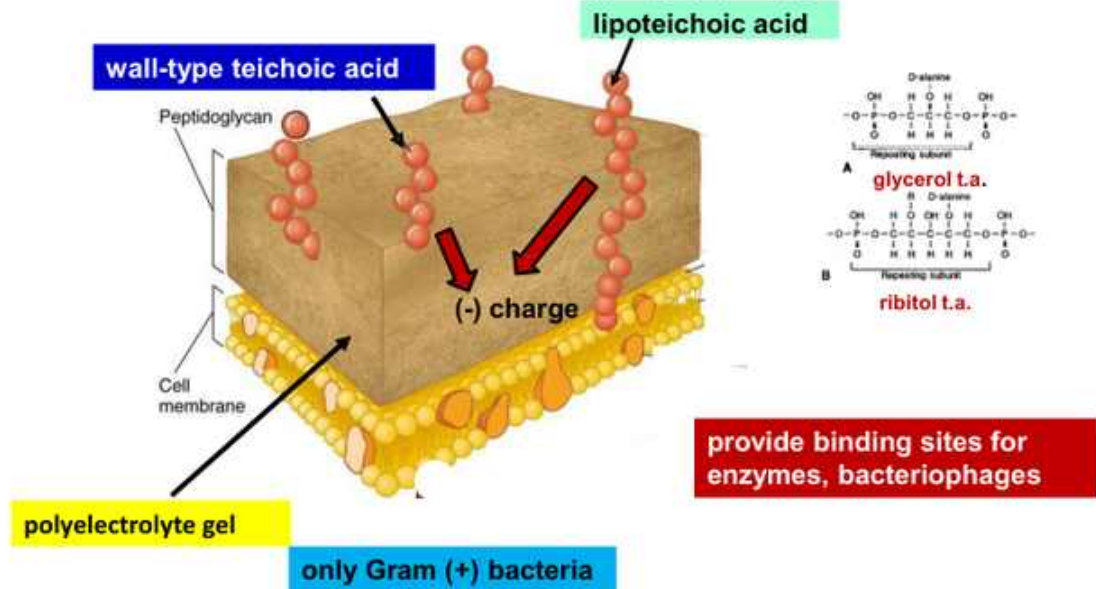
PEPTIDOGLYCAN FORMULA

NAG + NAM sugar backbone + peptide side chains/cross-linking. The deck lists L-alanine, glutamic acid, lysine, and D-alanine in the peptide component.

Handout-only drug connection: transpeptidase enzymes, also called **penicillin-binding proteins (PBPs)**, cross-link adjacent peptidoglycan strands. Beta-lactam antibiotics inhibit PBPs, weakening the wall and promoting bacterial lysis.

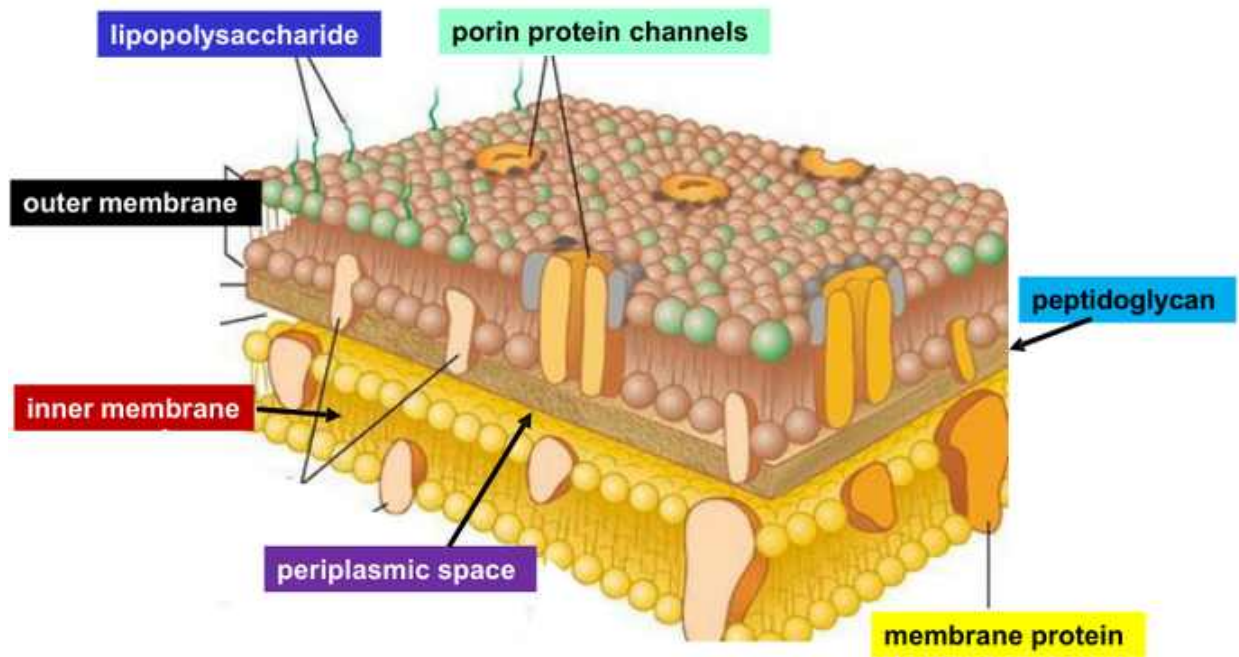
Teichoic Acids

regulate cation movement in & out
(ion homeostasis)



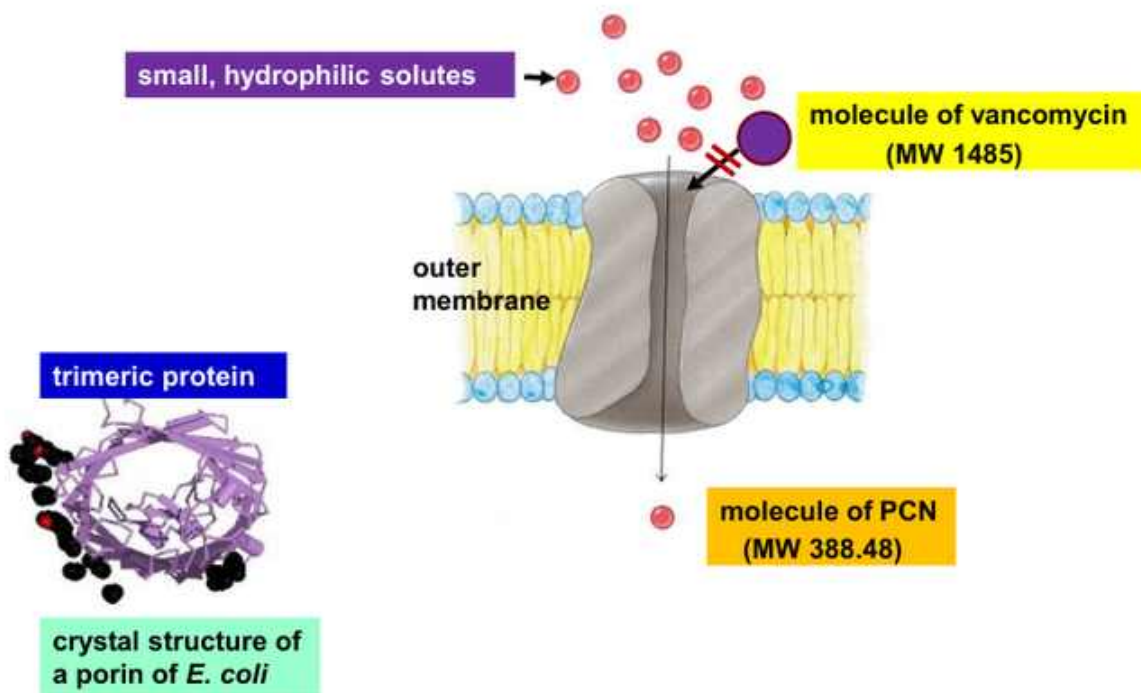
Slide 37. Teichoic acids are negatively charged and only in Gram-positive bacteria; the deck emphasizes ion homeostasis and binding/channel functions.

Gram-negative



Slide 39. Gram-negative envelope: outer membrane, LPS, porins, periplasmic space, thin peptidoglycan, and inner membrane.

Porin Proteins



Slide 40. Porins are trimeric outer-membrane channels that admit small hydrophilic solutes and can exclude larger molecules.

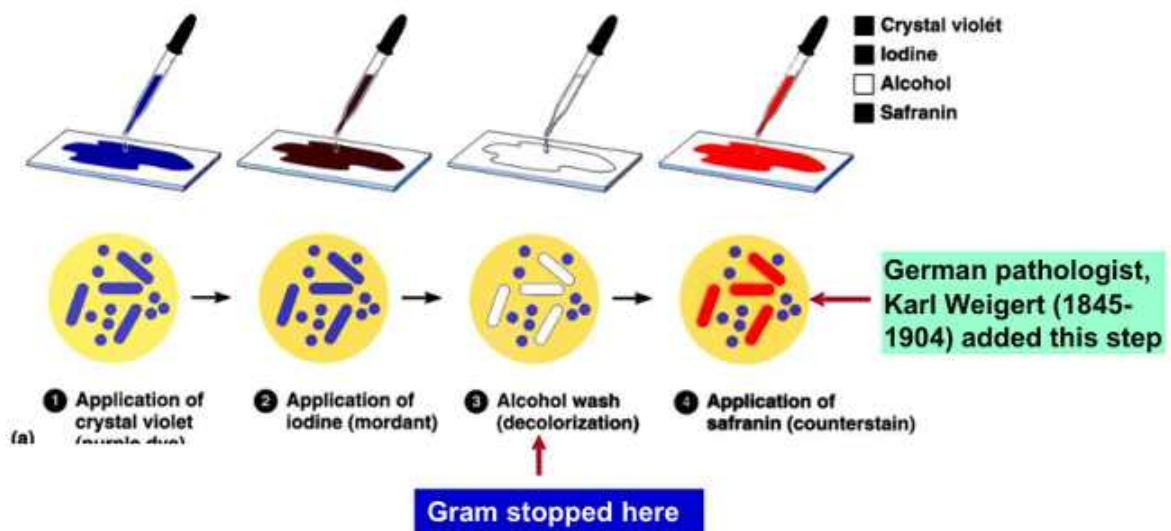
GRAM-NEGATIVE BARRIER LOGIC

Outer membrane + porins = an extra permeability barrier. The slide visually contrasts a smaller penicillin molecule with the much larger vancomycin molecule.

Handout-only LPS breakdown: Gram-negative LPS contains **O antigen** for serotyping, a **core polysaccharide**, and **Lipid A**, the toxic endotoxin moiety. The handout links Lipid A recognition through **TLR4** to IL-1, IL-6, and TNF-alpha release, producing fever, hypotension, and potentially DIC/septic shock. For Gram-positive organisms, the handout highlights **lipoteichoic acid** as an inflammatory PAMP.

Slides 44-54: Gram Stain Theory, Technique & Morphologic Reading

Gram's Staining



mordant: a substance capable of deepening the reaction of a specimen to a stain

Slide 45. Gram stain sequence: crystal violet → iodine mordant → alcohol/acetone decolorization → safranin counterstain.

GRAM STAIN ORDER

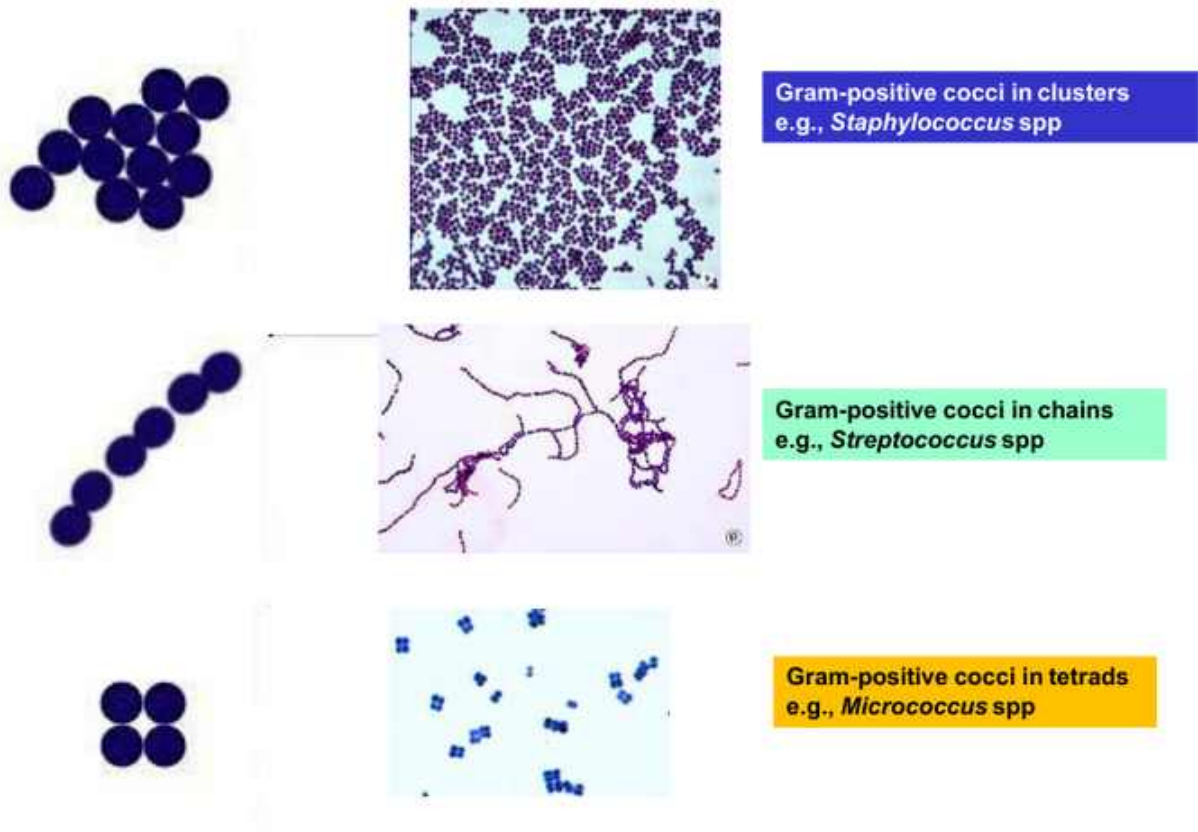
Crystal violet → iodine → decolorizer → safranin.

Step	Role
Crystal violet	Primary stain
Gram iodine	Mordant ; deepens/stabilizes the dye complex
Acetone-alcohol	Decolorizer ; key differential step
Safranin	Counterstain

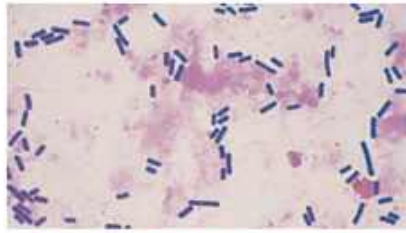
Ozzy checkpoint - click him once you can predict what over- or under-decolorization does to a smear.

Why Gram-positive stays purple: the deck's assessment item ties retention of the crystal-violet/iodine complex during decolorization to **greater peptidoglycan content**.

Technique trap: too little decolorizer can make everything look Gram-positive; too much decolorizer can make everything look Gram-negative. The deck tests these errors using neutrophil nuclear staining as a clue.



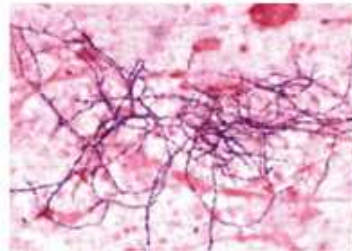
Slide 46. Gram-positive cocci morphology: clusters (*Staphylococcus*), chains (*Streptococcus*), and tetrads (*Micrococcus*).



**Thick Gram-positive bacilli,
e.g., *Clostridium* spp**

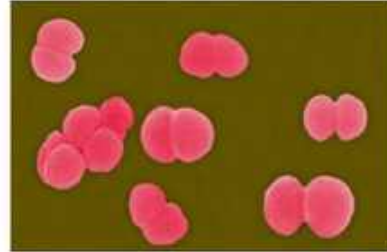
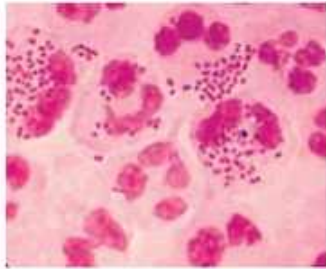


**Thin Gram-positive bacilli,
e.g., *Listeria* spp**



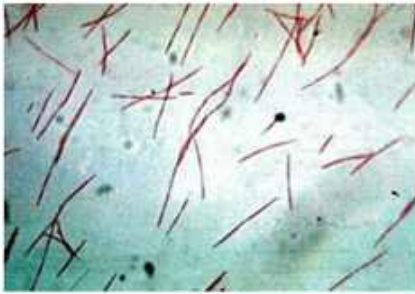
**Branched Gram-positive bacilli,
e.g., *Actinomyces* spp**

Slide 48. Gram-positive bacilli: thick *Clostridium*-type rods, thin *Listeria*-type rods, and branched *Actinomyces*-type forms.



***Neisseria gonorrhoeae*, urethral exudate,
Gram (-) diplococci**

Slide 49. *Neisseria gonorrhoeae* shown as Gram-negative diplococci in urethral exudate.



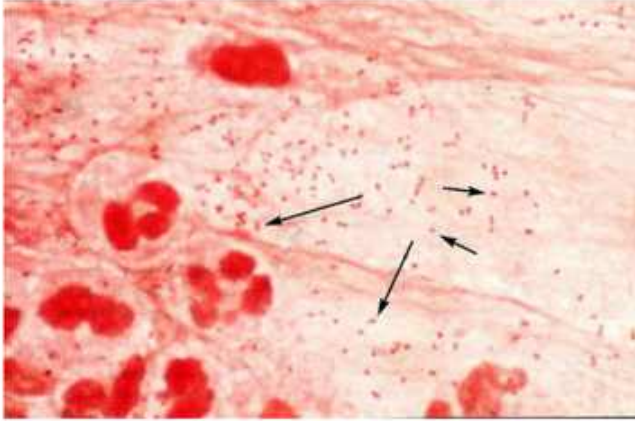
Elongate Gram-negative rods,
e.g., *Fusobacterium* spp



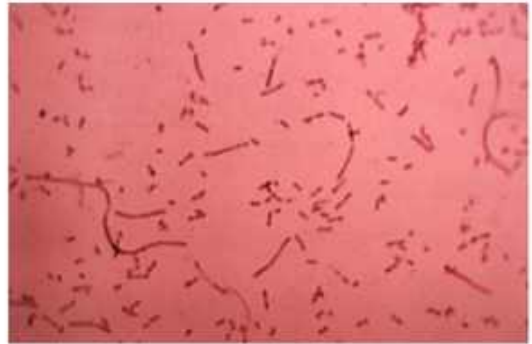
Shorter, thick (or thin) Gram-negative rods, e.g., *E. coli*,
Klebsiella pneumoniae, *Pseudomonas aeruginosa*

Slide 50. Gram-negative rod morphology: elongate *Fusobacterium* vs shorter rods such as *E. coli*, *Klebsiella*, and *Pseudomonas*.

Haemophilus influenzae



cocci



coccobacilli

Slide 54. *Haemophilus influenzae* demonstrates that organisms may appear coccoid or coccobacillary.

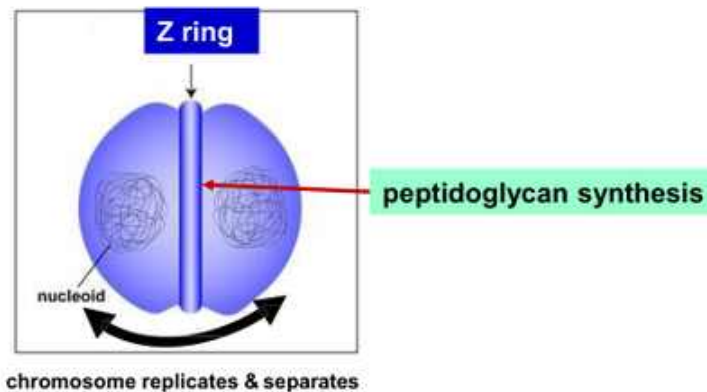
READ A GRAM STAIN IN TWO PASSES

1) Color: positive or negative. 2) Morphology/arrangement: cocci vs rods, chains vs clusters, diplococci, coccobacilli, branching.

Slides 36 & 43: Bacterial Division

Bacterial division

Fts proteins localize to division site (divisome)



Fts Z → polymerizes into tubulin → constricting Z ring

Fts = filamentation (temperature-sensitive)

Slide 36. Division-site biology: chromosome replication/separation, Fts proteins at the divisome, Z-ring formation, and new peptidoglycan synthesis.

FTSZ

FtsZ is the key Z-ring protein in the lecture. The Z ring constricts at the division site while new cell-wall material is synthesized.

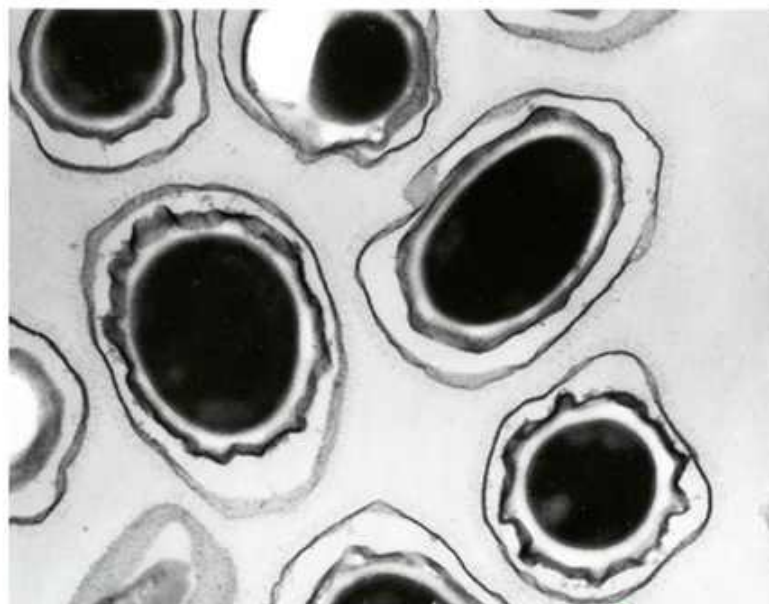
Deck wording: slide 36 describes “Fts Z → polymerizes into tubulin → constricting Z ring.” This companion preserves the slide’s tested association: **FtsZ = Z-ring / divisome / bacterial division.**

Binary fission: replicate the single bacterial chromosome → segregate copies → septum/cell-wall synthesis → divide into daughter cells.

Handout-only sequence detail: the cell elongates and replicates its circular chromosome, chromosome copies move apart, **FtsZ** assembles into the central Z-ring, the ring constricts with septum/cell-wall formation, and two daughter cells separate. The handout explicitly calls FtsZ a bacterial homolog of eukaryotic tubulin.

Slides 57-63: Bacterial Spores

Released spore can survive extreme heat, lack of water, toxic chemicals, radiation

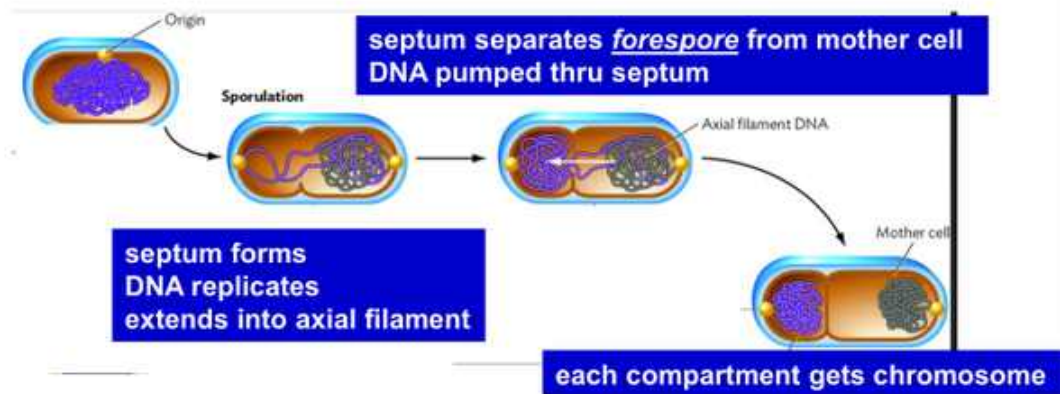


spore coat

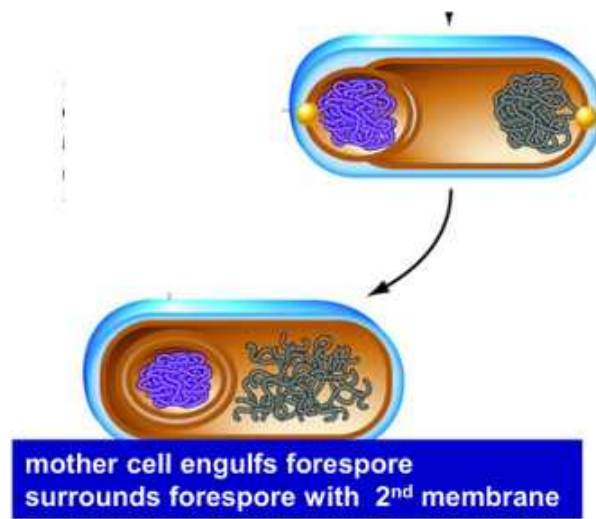
Slide 58. Released endospore survival message: spores tolerate extreme heat, lack of water, toxic chemicals, and radiation.

WHY SPORES MATTER

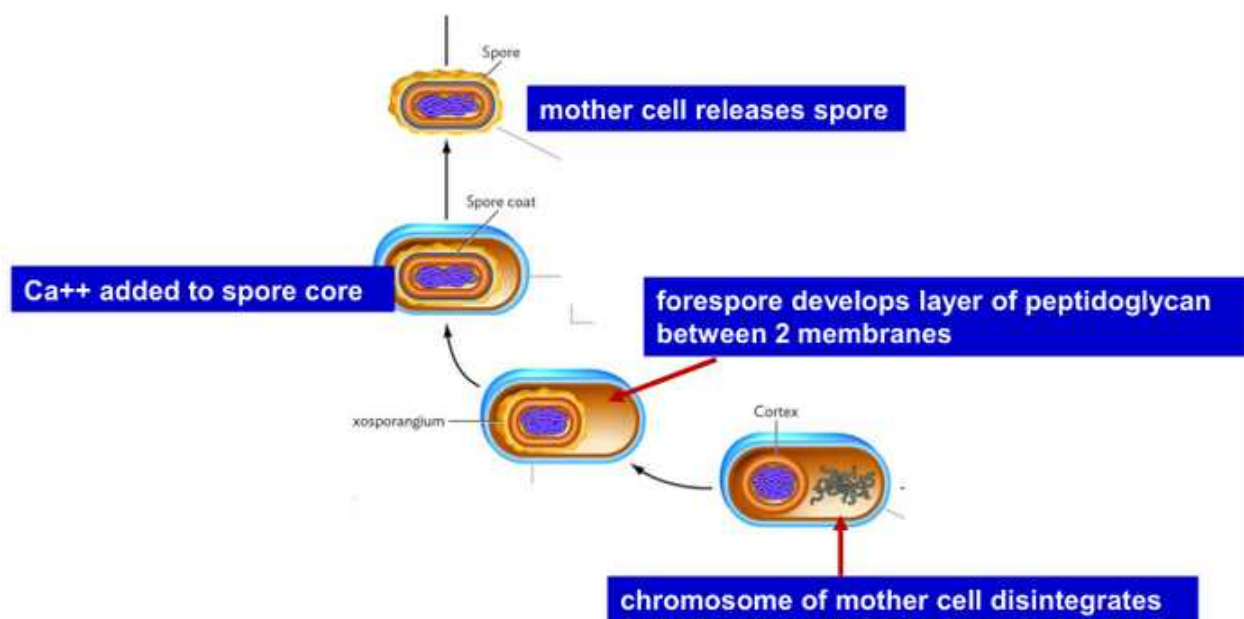
Endospores are survival structures, not a reproductive multiplication strategy. They allow persistence through otherwise lethal environmental stress.



Slide 60. Early sporulation: chromosome replicates and extends into an axial filament; an asymmetric septum separates forespore from mother cell.

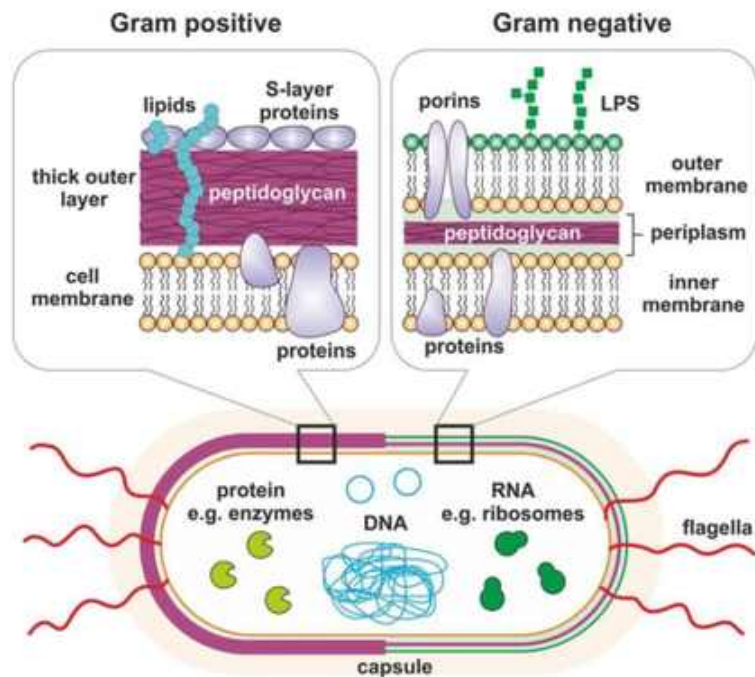


Slide 61. Mother cell engulfs the forespore, surrounding it with a second membrane.



Slide 62. Forespore develops peptidoglycan between the two membranes, calcium is added to the core, and the mother cell eventually releases the mature spore.

Summary



Slide 63. Faculty summary slide consolidates the spore-formation sequence.

Sporulation step	Deck sequence
1	DNA replication / axial filament
2	Asymmetric septum creates forespore + mother cell
3	Mother cell engulfs forespore → second membrane
4	Peptidoglycan layer develops between membranes
5	Calcium added to spore core; maturation
6	Mother cell releases mature spore

Handout-only clinical enrichment: endospores contain a protected core with **dipicolinic acid** and are built for survival, not reproduction. The handout emphasizes medically important spore formers in the **Bacillus** and **Clostridium** genera and uses reheated-rice *Bacillus cereus* food poisoning as a classic survival vignette.

High-Yield Vignette & Assessment Traps

Question stem	Best answer from this deck
Component in prokaryotic cells but not eukaryotic cells	Peptidoglycan
Gram-negative-only outer-membrane permeability structure	Porin protein
Gram-positive retains crystal violet after decolorization because of more...	Peptidoglycan
Gram-stain mordant	Gram iodine
Negatively charged Gram-positive polymer regulating cations	Teichoic acid
Key divisome/Z-ring protein	FtsZ
Pneumonia organism with no cell wall	Mycoplasma per the deck's question
Extrachromosomal circular resistance DNA	Plasmid
Essential peptidoglycan sugar	N-acetylglucosamine (NAG)
Pilin-containing adherence projections	Fimbriae
Flagellum part anchored to membrane	Basal body
Spirochete motility	Flagella beneath external sheath

Handout practice-vignette takeaways:

Handout stem	High-yield answer
Pink Gram-negative rod + profound septic shock	Lipid A of LPS
Walking pneumonia organism with no cell wall	Penicillin / beta-lactams ineffective
Prominent bacterial capsule / halo	Antiphagocytic virulence
Reheated fried rice, preformed toxin, Gram-positive rod	Endospore survival

Structure present in <i>E. coli</i> but not <i>S. aureus</i>	Outer membrane
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LAST 60 SECONDS

Gram+ = thick PG + teichoic acid. Gram- = thin PG + outer membrane/LPS/porins. Gram stain = CV → iodine → alcohol → safranin. FtsZ = division. NAG/NAM = peptidoglycan. Spores = extreme survival.

Source hierarchy: Fisher 78-slide current deck is primary. Official Week 8 LO sheet supplies the 6 objectives and ScholarRx brick. The user-supplied **Lecture 1: Bacterial Morphology and Structure** handout is secondary enrichment and is explicitly labeled when it adds clinical or board-style material beyond the current deck.